# Set Theory for Computer Science

## Week 4: Equivalence Relations

**Equivalence relations**

Equivalence classes

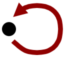
Equivalence relations v. partitions

Complete system of representatives

# Equivalence relations

An **equivalence relation** in $A$ is a relation of type $A \times A$ that is reflexive and transitive and symmetric

## Reminder

- reflexivity $x \, R \, x$ 
- transitivity $x \, R \, y \wedge y \, R \, z \to x \, R \, z$ 
- symmetry $x \, R \, y \to y \, R \, x$ 

# Example: congruence modulo an integer

## Congruence modulo $m$ in the integers

$$x \equiv y \pmod{m} \qquad \iff \qquad y - x \text{ is a multiple of } m$$

## Why an equivalence relation?

- reflexivity: $x - x = 0 \cdot m$
- symmetry: $y - x = p \cdot m \quad \iff \quad x - y = -p \cdot m$
- transitivity: $y - x = p \cdot m$ and $z - y = q \cdot m$

$$\Rightarrow \quad z - x = (z - y) + (y - x) = (p + q) \cdot m$$

# Example: congruence modulo an integer

### Congruence modulo $m$ in the integers

$$x \equiv y \pmod{m} \qquad \Longleftrightarrow \qquad y - x \text{ is a multiple of } m$$

### Examples

- Clocks and time: $16 \equiv 4 \pmod{12}$
- Weekdays: $11 \equiv 25 \pmod{7}$
- Binary arithmetic: $27 \equiv 283 \pmod{256}$

$$\begin{array}{r} 11001000 \\ 01010011 \\ \hline 00011011 \end{array} + \qquad \begin{array}{r} 200 \\ 83 \\ \hline 27 \end{array} +$$

# Example: logical equivalence

## Logical equivalence of propositional sentences

$\phi \equiv \psi \quad \Longleftrightarrow \quad$ $\phi$ and $\psi$ have the same truth value for every possible valuation

## Why an equivalence relation?

- reflexivity: $\phi$ always has same truth value as $\phi$
- symmetry: immediate from the definition
- transitivity: if $\phi$ and $\psi$ have the same truth value and $\psi$ and $\chi$ have the same truth value, then $\phi$ and $\chi$ have the same truth value

# Example: logical equivalence

## Logical equivalence of propositional sentences

$\phi \equiv \psi \quad \Longleftrightarrow \quad$ $\phi$ and $\psi$ have the same truth value for every possible valuation

## Examples

- De Morgan:
$$\begin{aligned} \neg(p \wedge q) &\equiv \neg p \vee \neg q \\ \neg(p \vee q) &\equiv \neg p \wedge \neg q \end{aligned}$$
- distributivity:
$$\begin{aligned} p \wedge (q \vee r) &\equiv (p \wedge q) \vee (p \wedge r) \\ p \vee (q \wedge r) &\equiv (p \vee q) \wedge (p \vee r) \end{aligned}$$
- contradictions:
$$\neg p \wedge p \equiv \neg q \wedge q$$

**Exercise**

In $A := \{\mathrm{a}, \mathrm{b}, \mathrm{c}\}^2 = \{\mathrm{aa}, \mathrm{ab}, \mathrm{ac}, \mathrm{ba}, \mathrm{bb}, \mathrm{bc}, \mathrm{ca}, \mathrm{cb}, \mathrm{cc}\}$, define

$$x \mathrel{R} y \iff x \text{ and } y \text{ have the same number of a's}$$
$$x \mathrel{Q} y \iff x \text{ and } y \text{ start with the same letter}$$

(a) Is $R$ an equivalence relation in $A$? And what about $Q$?

(b) List all $x \in A$ such that $x \mathrel{R} \mathrm{ab}$ holds.

(c) List all $x \in A$ such that $x \mathrel{Q} \mathrm{ab}$ holds.

**Exercise**

In $A := \{a, b, c\}^2 = \{aa, ab, ac, ba, bb, bc, ca, cb, cc\}$, define

$$x \, R \, y \quad \Longleftrightarrow \quad x \text{ and } y \text{ have the same number of a's}$$
$$x \, Q \, y \quad \Longleftrightarrow \quad x \text{ and } y \text{ start with the same letter}$$

(a) Is $R$ an equivalence relation in $A$? And what about $Q$?

both are equivalence relations

(b) List all $x \in A$ such that $x \, R$ ab holds.

$x \, R$ ab holds for all $x \in \{ab, ac, ba, ca\}$

(c) List all $x \in A$ such that $x \, Q$ ab holds.

$x \, Q$ ab holds for all $x \in \{aa, ab, ac\}$

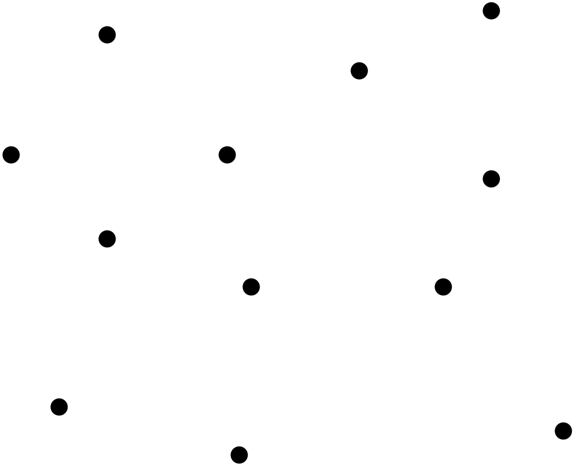
Equivalence relations

**Equivalence classes**

Equivalence relations v. partitions

Complete system of representatives

# What does an equivalence relation look like?



✗ reflexive
✓ symmetric
✓ transitive

✗ equiv. rel.

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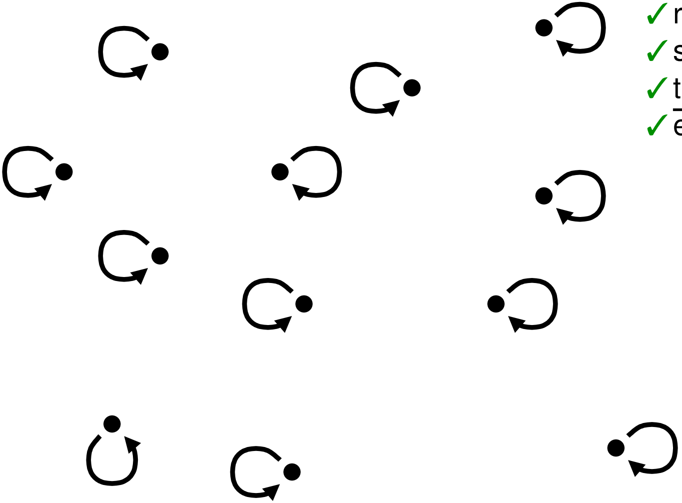

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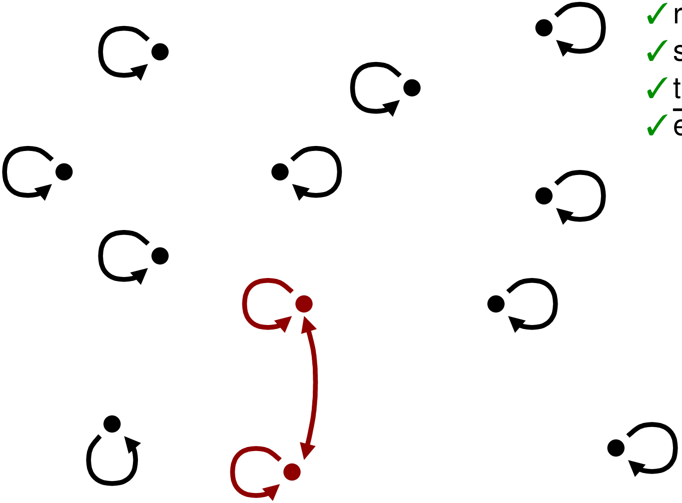

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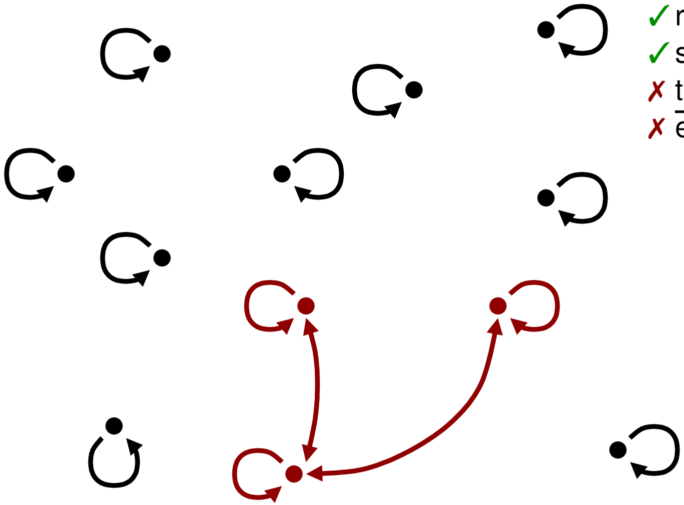

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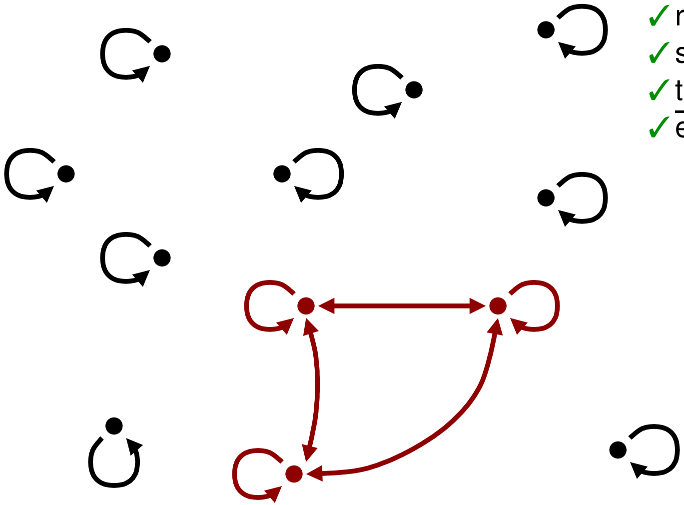

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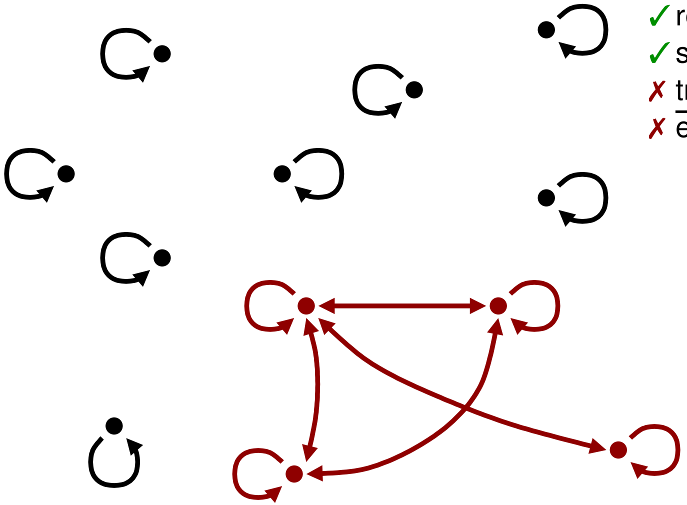

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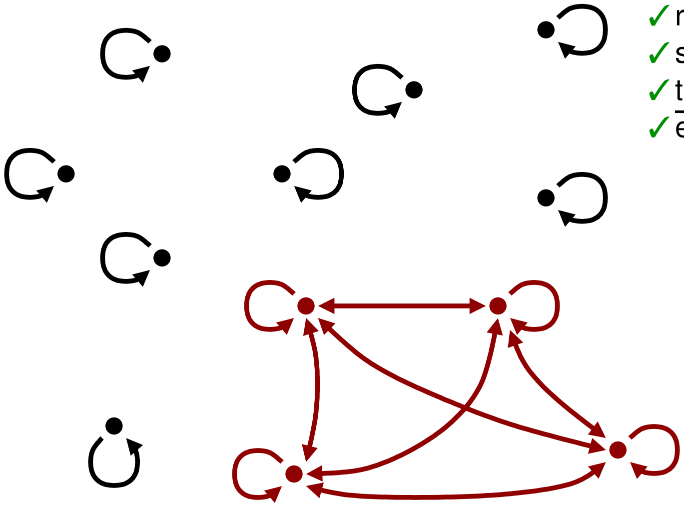

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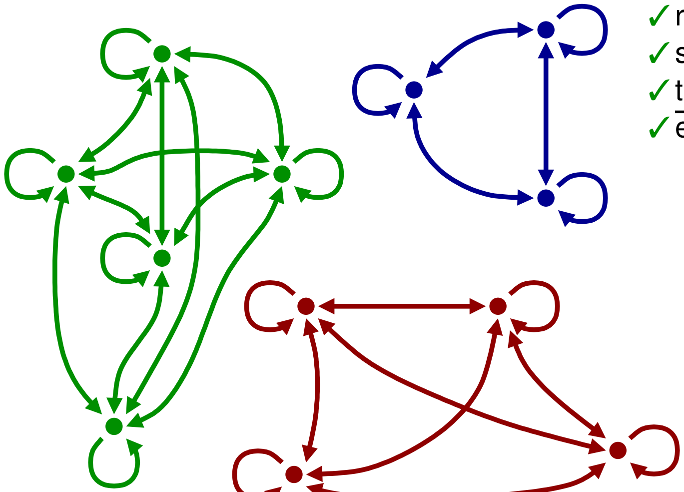

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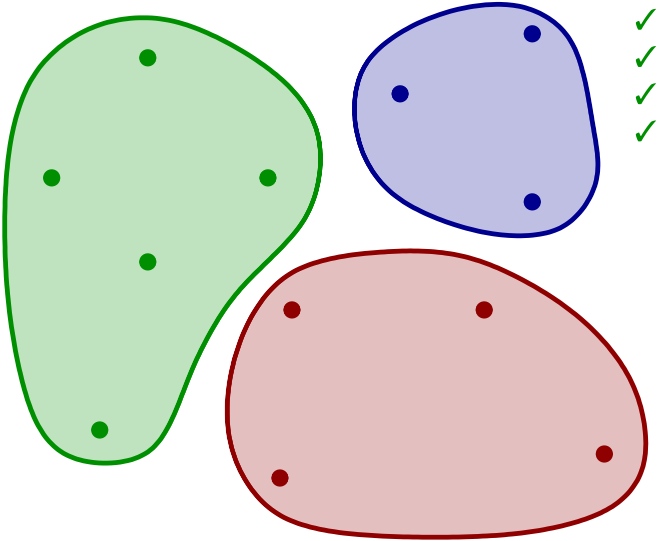

✓ reflexive
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$\Downarrow$

**equivalence classes**

# Equivalence classes

Let $\equiv$ be an equivalence relation in the set $A$

The **equivalence class** of an element $a \in A$ is the set

$$[a] := \{x \in A : x \equiv a\}$$

- all elements of an equivalence class relate to each other
- elements of different equivalence classes are unrelated

**Theorem**
The equivalence classes together form a partition of the set $A$

# Example: 3-bit numbers

**Example**

In the set $A := \{0, 1\}^3$ of 3-bit numbers, define

$$x \equiv y \qquad \Longleftrightarrow \qquad x \text{ and } y \text{ contain the same number of 0's}$$

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$$x \equiv y \qquad \Longleftrightarrow \qquad x \text{ and } y \text{ contain the same number of 0's}$$

**Equivalence classes**

$$\begin{array}{ll}
[000] = \{000\} & [011] = \{011, 101, 110\} \\
[001] = \{001, 010, 100\} & [101] = \{011, 101, 110\} \\
[010] = \{001, 010, 100\} & [110] = \{011, 101, 110\} \\
[100] = \{001, 010, 100\} & [111] = \{111\}
\end{array}$$

**NB** there are *four* equivalence classes (two have three names)

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**NB** there are *four* equivalence classes (two have three names)

$$\{[a] \colon a \in A\} = \{[000], [001], [011], [111]\} \text{ is a partition of } A$$

**Example**

Consider congruence modulo 3 in $\mathbb{N}$

# Example: congruence modulo an integer

## Example

Consider congruence modulo 3 in $\mathbb{N}$

## Equivalence classes

$$
\begin{aligned}
[0] &= \{n \in \mathbb{N} \colon n - 0 \text{ is a multiple of } 3\} = \{3n + 0 \colon n \in \mathbb{N}\} \\
[1] &= \{n \in \mathbb{N} \colon n - 1 \text{ is a multiple of } 3\} = \{3n + 1 \colon n \in \mathbb{N}\} \\
[2] &= \{n \in \mathbb{N} \colon n - 2 \text{ is a multiple of } 3\} = \{3n + 2 \colon n \in \mathbb{N}\}
\end{aligned}
$$

# Example: congruence modulo an integer

**Example**

Consider congruence modulo 3 in $\mathbb{N}$

**Equivalence classes**

$$[0] = \{n \in \mathbb{N} \colon n - 0 \text{ is a multiple of } 3\} = \{3n + 0 \colon n \in \mathbb{N}\}$$
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$\{[n] \colon n \in \mathbb{N}\} = \{[0], [1], [2]\}$ is a partition of $\mathbb{N}$

**Exercise**

In $A := \{a, b, c\}^2 = \{aa, ab, ac, ba, bb, bc, ca, cb, cc\}$, define

$$x\,R\,y \iff x \text{ and } y \text{ have the same number of a's}$$
$$x\,Q\,y \iff x \text{ and } y \text{ start with the same letter}$$

(a) Determine all equivalence classes of the relation $R$, and write out the partition of $A$ formed by these equivalence classes.

(b) Do the same for the relation $Q$.

**Exercise**

In $A := \{\mathrm{a}, \mathrm{b}, \mathrm{c}\}^2 = \{\mathrm{aa}, \mathrm{ab}, \mathrm{ac}, \mathrm{ba}, \mathrm{bb}, \mathrm{bc}, \mathrm{ca}, \mathrm{cb}, \mathrm{cc}\}$, define

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(a) Determine all equivalence classes of the relation $R$, and write out the partition of $A$ formed by these equivalence classes.

$$\{\{\mathrm{aa}\}, \{\mathrm{ab}, \mathrm{ac}, \mathrm{ba}, \mathrm{ca}\}, \{\mathrm{bb}, \mathrm{bc}, \mathrm{cb}, \mathrm{cc}\}\}$$

(b) Do the same for the relation $Q$.

$$\{\{\mathrm{aa}, \mathrm{ab}, \mathrm{ac}\}, \{\mathrm{ba}, \mathrm{bb}, \mathrm{bc}\}, \{\mathrm{ca}, \mathrm{cb}, \mathrm{cc}\}\}$$

# break

Equivalence relations

Equivalence classes

**Equivalence relations v. partitions**

Complete system of representatives

# From equivalence relation to partition

equivalence relation $\equiv$ in $A$

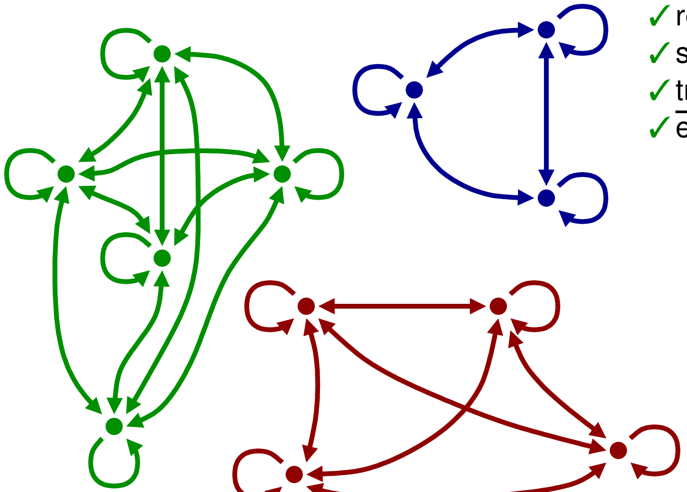
$$\Downarrow$$

equivalence classes $[a]$

$$\Downarrow$$

partition $\{[a] : a \in A\}$ of $A$

# From equivalence relation to partition



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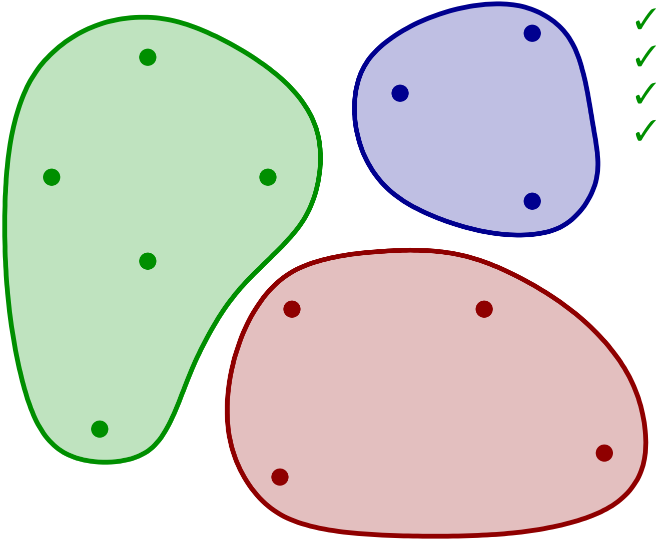

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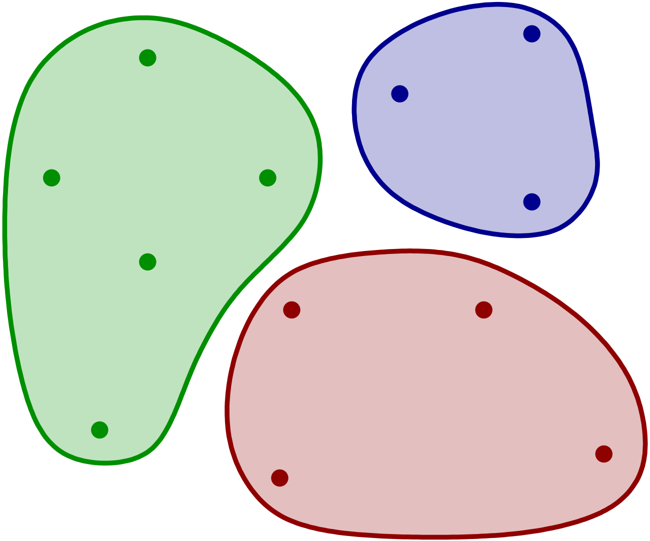
$\Downarrow$

equivalence classes

$\Downarrow$

partition

# From partition to equivalence relation

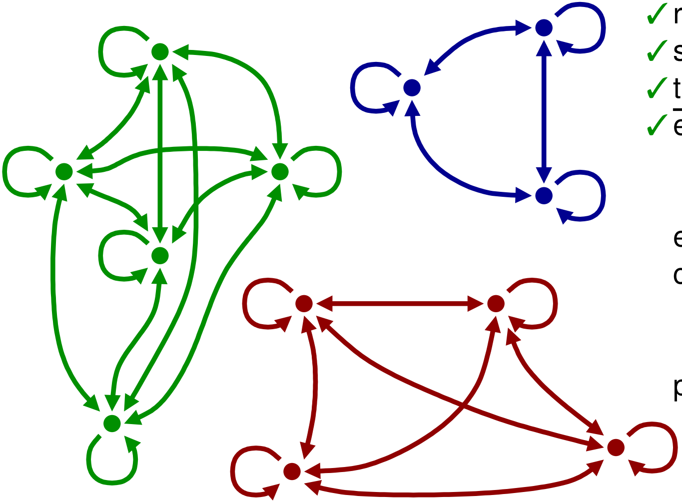


equivalence
classes

$\Uparrow$

partition

# From partition to equivalence relation



✓ reflexive
✓ symmetric
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---

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$\Uparrow$

equivalence classes

$\Uparrow$

partition

# From partition to equivalence relation

partition $P = \{P_1, P_2, \dots\}$ of $A$

$$\Downarrow$$

define $\equiv$ in $A$ by

$$x \equiv y \iff x \text{ and } y \text{ are in the same part}$$

$$\Downarrow$$

$\equiv$ is an equivalence relation
and $\{[a] : a \in A\} = P$

## Even and odd integers

Partition $\mathbb{N}$ into **even** and **odd** integers

## Equivalence relation

The corresponding equivalence relation is congruence modulo 2:

$$x \equiv y \iff x - y \text{ is a multiple of } 2$$


## Equivalence classes

$[0] = \{n \in \mathbb{N} : n \text{ even}\}$
$[1] = \{n \in \mathbb{N} : n \text{ odd}\}$

Equivalence relations

Equivalence classes

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**Complete system of representatives**

# Complete system of representatives

A **complete system of representatives** for $\equiv$ in $A$ is a set $S \subseteq A$ that contains *precisely one* element from each equivalence class

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### Example

Consider congruence modulo 3 in $\mathbb{N}$

### Partition of equivalence classes

$$\{[n] \colon n \in \mathbb{N}\} = \{[0], [1], [2]\}$$


### Complete system of representatives

$S = \{0, 1, 2\}$ (for example)

# Complete system of representatives

A **complete system of representatives** for $\equiv$ in $A$ is a set $S \subseteq A$ that contains *precisely one* element from each equivalence class

**Example**

In the set $A := \{0, 1\}^3$ of 3-bit numbers, consider

$$x \equiv y \qquad \Longleftrightarrow \qquad x \text{ and } y \text{ have the same number of 0's}$$

**Partition of equivalence classes**

$$\{[a] : a \in A\} = \{[000], [001], [011], [111]\}$$

**Complete system of representatives**

$$S = \{000, 001, 011, 111\} \qquad\qquad \text{(for example)}$$

# Complete systems of representatives: characterization

**Theorem**

A subset $S$ of $A$ is a complete system of representatives for an equivalence relation $\equiv$ in $A$ if and only if

1. every $a \in A$ is equivalent to some $s \in S$
2. two different elements of $S$ are not equivalent

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**Proof ($\Rightarrow$ direction)**

1. let $a \in A$ $\quad\Rightarrow\quad$ $S$ contains an element $s \in [a]$
   $\quad\Rightarrow\quad$ $a \equiv s$ for some $s \in S$

2. let $s, s' \in S$ $\quad\Rightarrow\quad$ if $s \neq s'$ then $[s] \neq [s']$
   $\quad\Rightarrow\quad$ if $s \neq s'$ then $s \not\equiv s'$

# Complete systems of representatives: characterization

**Theorem**

A subset $S$ of $A$ is a complete system of representatives for an equivalence relation $\equiv$ in $A$ if and only if

**1.** every $a \in A$ is equivalent to some $s \in S$

**2.** two different elements of $S$ are not equivalent

**Proof ($\Leftarrow$ direction)**

**1.** let $a \in A$ $\quad \Rightarrow \quad a \equiv s$ for some $s \in S$

$\Rightarrow \quad S$ contains an element $s$ from $[a]$

**2.** $\quad \Rightarrow \quad$ if $s' \in S$ and $s \neq s'$ then $s \not\equiv s'$

$\Rightarrow \quad$ if $s' \in S$ and $s \neq s'$ then $s' \notin [s] = [a]$

$\Rightarrow \quad S$ contains exactly one element from $[a]$

**Exercise**

In $A := \{a, b, c\}^2 = \{aa, ab, ac, ba, bb, bc, ca, cb, cc\}$, define

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(a) Give a complete system of representatives for $R$.

(b) Give a complete system of representatives for $Q$.

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(a) Give a complete system of representatives for $R$.

e.g. $\{aa, ab, bb\}$

(b) Give a complete system of representatives for $Q$.

e.g. $\{aa, bb, cc\}$